STAT 387 **HOMEWORK # 2 KEY** Winter 2026

INTRODUCTION TO STATISTICAL LEARNING

Conceptual

1. 4 points

KNN refers to a non-parametric method that can be used for classification or regression.

- In the case of classification, a KNN classifier identifies the nearest K points to the observation x_0 to be classified. It then estimates the probability of x_0 belonging to a specific class based on the fraction of that class amongst all the selected K points. Finally, the point is classified as belonging to the class with the highest probability. The KNN classifier provides a qualitative response.
- KNN regression is similar to classification in that to also uses the nearest K points to x_0. However, unlike a classifier it provides a quantitative prediction by averaging the responses of K points to estimate the $f(x_0)$

2. 6 points

Let Y represent the response (starting salary after graduation)

(a) (**3 points**)

Original model:

$$\hat{Y} = 50 + 20X_1 + 0.07X_2 + 35X_3 + 0.01X_4 - 10X_5$$

Level: College (1)

$$\hat{Y} = 50 + 20X_1 + 0.07X_2 + 35 + 0.01X_4 - 10X_1 \qquad (1)$$

Level: High school (0)

$$\hat{Y} = 50 + 20X_1 + 0.07X_2 + 0.01X_4 \qquad (2)$$

Difference between (1) and (2) $:= 35 - 10X_1$

If Level: High School = Level: College, then

$$35 - 10X_1 = 0 \implies X_1 = 3.5$$

But if $X_1 > 3.5$ (GPA is high enough, then high school graduates earn more on average than college graduates.

(iii) Because once the GPA is high enough, high school graduates earn more on average.

(b) (**3 points**)

With Level = 1, IQ = 110, GPA = 4

$$\hat{Y} = 50 + 20(4) + 0.07(110) + 35 + 0.01(4 * 110) - 10(4) = 137.1$$

This implies starting salary after graduation is $ 137,100

Applied

3. **7 points**

a.

From the regression fit we obtain the following results:

```
Call:
lm(formula = mpg ~ horsepower, data = dataA)

Residuals:
     Min       1Q   Median       3Q      Max 
-13.5710  -3.2592  -0.3435   2.7630  16.9240 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 39.935861   0.717499   55.66   <2e-16 ***
horsepower  -0.157845   0.006446  -24.49   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.906 on 390 degrees of freedom
Multiple R-squared:  0.6059,	Adjusted R-squared:  0.6049 
F-statistic: 599.7 on 1 and 390 DF,  p-value: < 2.2e-16
```

(**2 points**)

Comments:

i) From the above output, there is strong evidence of a relationship between the predictor (horsepower) and the response (mpg), and this is seen in the p-value for horsepower, [which is $< 2\times 10^{-16}$] which is less than 5% significance level.

ii) The R^2 statistic is approximately 0.61, and this means about 60% of variance in mpg can be explained by horsepower in this model. The relationship between mpg and horsepower is reasonably strong.

iii) mpg has a negative linear relationship with horsepower. For every unit increase in horsepower, the mpg decreases by 0.158mpg.
(**1 point**)

iv) From our regression fit, the estimated model is:

$$\widehat{mpg} = 39.935861 - 0.157845 * horsepower$$

Therefore, the predicted mpg with horsepower = 98 is

$$\widehat{mpg} = 39.935861 - 0.157845(98) = \mathbf{24.46708}\ mpg$$

The 95% confidence interval is:

[23.97308, 24.96108]

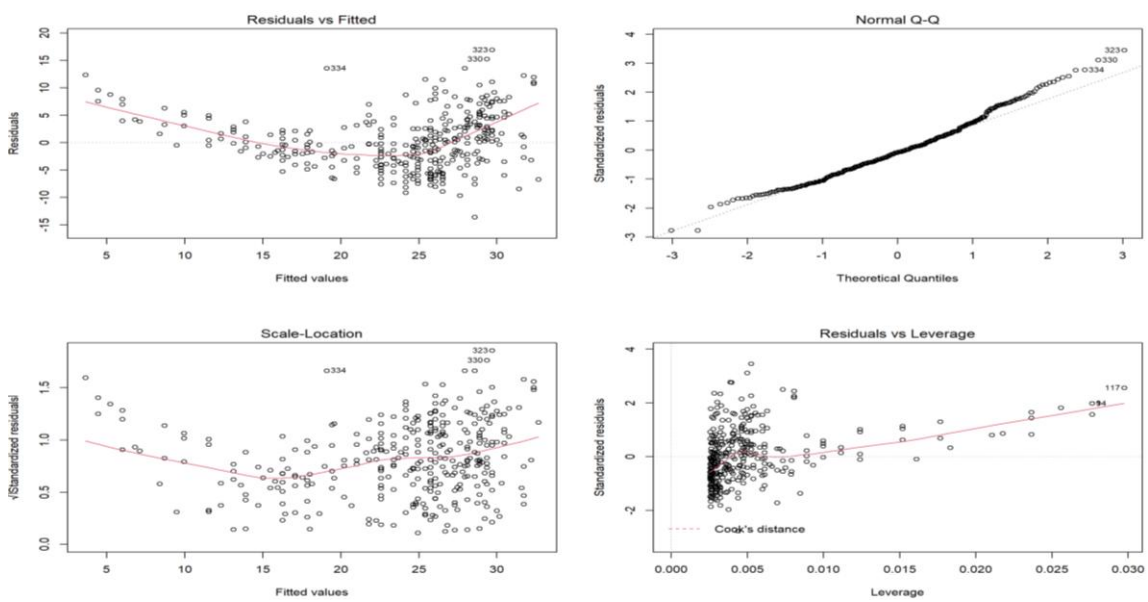
The 95% prediction interval is:

[14.8094, 34.12476]

(**1 point**)

b. (**1 point**)

c.



(**1 point**)

Comments:

- The residuals vs fitted chart shows a slight U-shaped pattern, and this indicates non-linearity in the data.
- The residuals vs leverage chart shows that some observations have high leverage.
- The scale-location chart shows some possible outliers. We can confirm by using studentized residuals to find observations with values greater than 3. (**1 point**)

4. **8 points**

a) (**1 point**)
The following are results from the multiple regression fit:

```
Call:
lm(formula = Sales ~ Price + Urban + US, data = dataB)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.9206 -1.6220 -0.0564  1.5786  7.0581 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 13.043469   0.651012  20.036  < 2e-16 ***
Price       -0.054459   0.005242 -10.389  < 2e-16 ***
UrbanYes    -0.021916   0.271650  -0.081    0.936    
USYes        1.200573   0.259042   4.635 4.86e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.472 on 396 degrees of freedom
Multiple R-squared:  0.2393,	Adjusted R-squared:  0.2335 
F-statistic: 41.52 on 3 and 396 DF,  p-value: < 2.2e-16
```

b) (**2 points**)

- The intercept represents the number of car seats sold on average when all other predictors are disregarded/ not present. Meaning with no predictor influence, Sales will be roughly 13043 (13.043 x 1000) seats.
- The **Price** coefficient is negative and so this means that Sales will drop by roughly 54 seats (0.054 x 1000) for every unit ($1) increase in price.
- With coefficient estimated at about -0.02, we can say an Urban store location **adversely impacts** Sales, relative to a store location that is not Urban.
- With a coefficient of about 1.20, it appears that a US-based store is associated with **high** Sales (relative to a store outside of the US).

c) (**1 point**)
From our regression fit, the estimated model is:

$$\widehat{Sales} = 13.043469 - 0.054459 * Price - 0.021916 * Urban(Yes) + 1.200573 * US(Yes)$$

d) (**2 points**)
Based on the p-values for the coefficients estimates above, we will reject the null hypotheses for **Price.** Although the p-value for **US** (Yes) is less than 5%, and then for **Urban** (Yes) the value is > 5%, we can properly comment on the significance of these categorical predictors by running Anova tests on respective reduced models.

e) (**1 point**)
The reduced fitted model produces the following results:

```
Call:
lm(formula = Sales ~ Price + US, data = dataB)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.9269 -1.6286 -0.0574  1.5766  7.0515 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 13.03079    0.63098  20.652  < 2e-16 ***
Price       -0.05448    0.00523 -10.416  < 2e-16 ***
USYes        1.19964    0.25846   4.641 4.71e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.469 on 397 degrees of freedom
Multiple R-squared:  0.2393,	Adjusted R-squared:  0.2354 
F-statistic: 62.43 on 2 and 397 DF,  p-value: < 2.2e-16
```

Therefore, our reduced regression model is:

$$\widehat{Sales} = 13.03079 - 0.05448 * Price + 1.19964 * US(Yes)$$

f) (**1 point**)
We decide based on the following test:
H_0: Reduced model is better, that is Coefficient for Urban $= 0$
H_1: Full model is better

```
Analysis of Variance Table

Model 1: Sales ~ Price + US
Model 2: Sales ~ Price + Urban + US
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    397 2420.9                           
2    396 2420.8  1   0.03979 0.0065 0.9357
```

With the obtained p-value (0.9357, which is > 5%), we fail to reject the null hypothesis and conclude that the reduced model is a better fit.

5. **5 points**

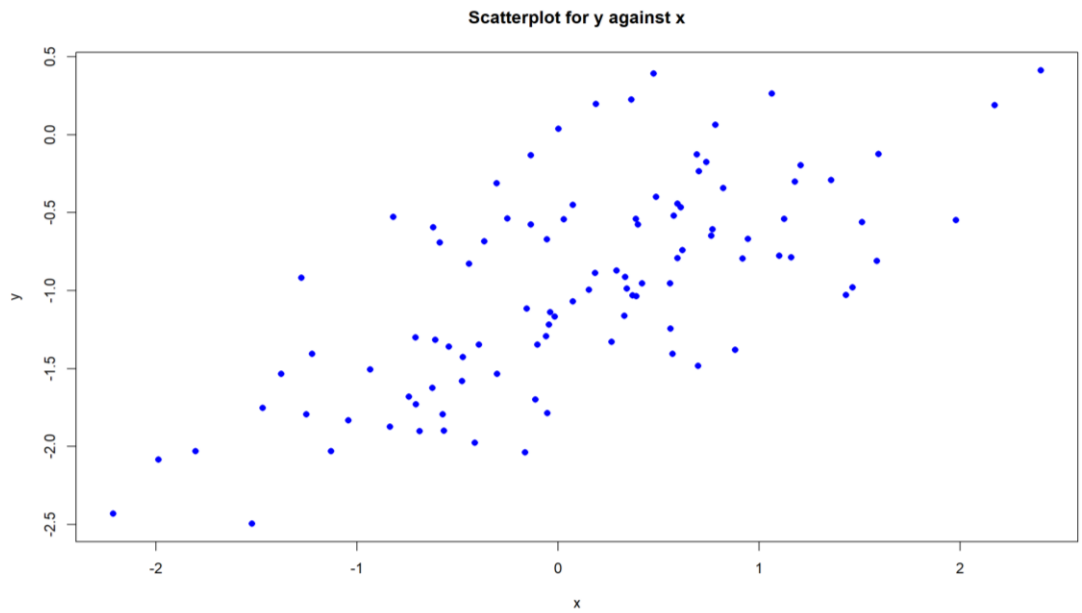
a) (**2 points**)
-0.6264538 0.1836433 -0.8356286 1.5952808 0.3295078 -0.8204684

b) (**1 point**)
-0.31018334 0.02105794 -0.45546082 0.07901439 -0.32729232 0.88364363

c) (**1 point**)
-1.6234102 -0.8871204 -1.8732751 -0.1233452 -1.1625384 -0.5265906

Length of Y is 100. $\beta_0 = -1$ and $\beta_1 = 0.5$ (**1 point**)

d) (**1 point**)



Comments:

A positive linear relationship appears to exist between x and y, with some added spread/variance which we suspect is introduced by the error term (eps).

e)

```
Call:
lm(formula = y ~ x)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.93842 -0.30688 -0.06975  0.26970  1.17309 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.01885    0.04849 -21.010  < 2e-16 ***
x            0.49947    0.05386   9.273 4.58e-15 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.4814 on 98 degrees of freedom
Multiple R-squared:  0.4674,	Adjusted R-squared:  0.4619 
F-statistic: 85.99 on 1 and 98 DF,  p-value: 4.583e-15
```

(**1 point**)

Comments: (**2 points**)

- Model obtained will read as: $\hat{y} = -1.01885 + 0.49947 * x$
- p-values for each estimated regression coefficient in this model and for the overall F-statistic are less than 0.05. Therefore, we reject the null hypothesis that $H_0: \beta_1 = 0$.
- Also, the regression coefficient estimates ($\widehat{\beta_0} = -1.01885$; $\widehat{\beta_1} = 0.49947$) are very close to the true values: ($\beta_0 = -1$ and $\beta_1 = 0.5$).

R Codes

Question 3: **2 points**

```r
rm(list = ls())

library(ISLR2)


dataA = ISLR2::Auto

# Question 3 #
#=== (a) ===#

fita = lm(mpg~horsepower, data = dataA)
summary(fita)

# (iv) #
pred98 = predict(fita, data.frame(horsepower = c(98)))
pred98

# Confidence interval #
pred95CI = predict(fita, data.frame(horsepower = c(98)), interval = "confidence")
pred95CI

# Prediction interval #

pred95PI = predict(fita, data.frame(horsepower = c(98)), interval = "prediction")
pred95PI


#=== (b) ===#
plot(mpg~horsepower, data = dataA, pch = 16, main = "Scatter plot vs Regression fit for MPG~Horsepower")
abline(fita, lwd = 3, col = "red")


#=== (c) ===#
par(mfrow = c(2,2))

plot(fita)

par(mfrow = c(1,1))
```

Question 4: **2 points**

```r
rm(list = ls())

library(ISLR2)


dataB = ISLR2::Carseats

# Question 4 #
#=== (a) ===#

fitm = lm(Sales ~ Price + Urban + US, data = dataB)
summary(fitm)


# (e) #

fitmred = lm(Sales ~ Price + US, data = dataB)
summary(fitmred)

# (f) #
anova(fitmred, fitm)
```

Question 5: **2 points**

```r
rm(list = ls())

set.seed(1)

# (a) #
x = rnorm(100)
head(x)

# (b) #
eps = rnorm(100,0,sqrt(0.25))
head(eps)

# (c) #
y = -1 + 0.5*x + eps
head(y)

# (d) #
plot(x, y, pch = 16, col = "blue", main = "Scatterplot of y against x")

# (e) #
fit.lm = lm(y~x)
summary(fit.lm)
```